UNIT–2 AGILE MINI-PROJECT

Student Record and Academic Management Platform

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Agile Project Planning and Documentation Report

Project Information	Details
Project Type	Academic Mini-Project
Approach	Agile / Scrum
Scope	Student records, courses, attendance, academic performance, security, notifications and reporting
Primary Users	Students, Faculty Members and authorized institutional users
Unit	Unit–2 Agile Mini-Project

Prepared from the supplied Unit–2 mini-project brief.

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1. Project Overview

The Student Record and Academic Management Platform is a proposed secure, user-friendly web-based application for managing core academic information. The Unit–2 brief asks the team to translate Unit–1 product-discovery material into Agile deliverables: Product Vision, Scrum roles, Epics, User Stories, Acceptance Criteria, Product Backlog, Story Points, Sprint Planning, Scrum Board, Scrum ceremonies, framework comparison and a release roadmap.

1.1 Objectives

Centralize student academic records.

Support student registration and profile management.

Manage courses and student-course assignments.

Record and report attendance.

Maintain marks and academic performance information.

Provide login, logout, password reset and role-based access control.

Generate student, attendance, performance and course reports.

Provide attendance, examination and result notifications.

Demonstrate incremental delivery using Agile/Scrum.

2. Business Problem and Product Vision

The team must first review the Product Vision, Mission, stakeholders, Business Requirements, Functional and Non-functional Requirements, User Personas, MVP and Product Roadmap from Unit–1. The purpose is to understand the intended product before converting it into Agile deliverables.

2.1 Business Need

A unified academic platform can provide a structured way to maintain student information, attendance, courses, marks and reports, while supporting timely access to academic information and notifications.

2.2 Product Vision

> **Develop a secure and user-friendly Student Record and Academic Management Platform that enables educational institutions to efficiently manage student records, attendance, academic performance, course registration, and reporting through a scalable web-based application.**

This vision is the reference point for prioritizing features and evaluating whether Sprint work contributes to the intended product.

3. Agile Product Vision

The Agile Product Vision identifies the product objective, primary users, business benefits, measurable success criteria, constraints and expected software capabilities.

Vision Element	Definition
Business benefits	Centralized records, easier attendance/ performance management, faster reporting and improved information access.
Success criteria	Priority stories are accepted against measurable Acceptance Criteria and delivered through planned Sprints.
Constraints	Limited project time, team capacity, implementation complexity and dependencies.
Expected capabilities	Student, course, attendance, marks, authentication, notification and reporting functions.

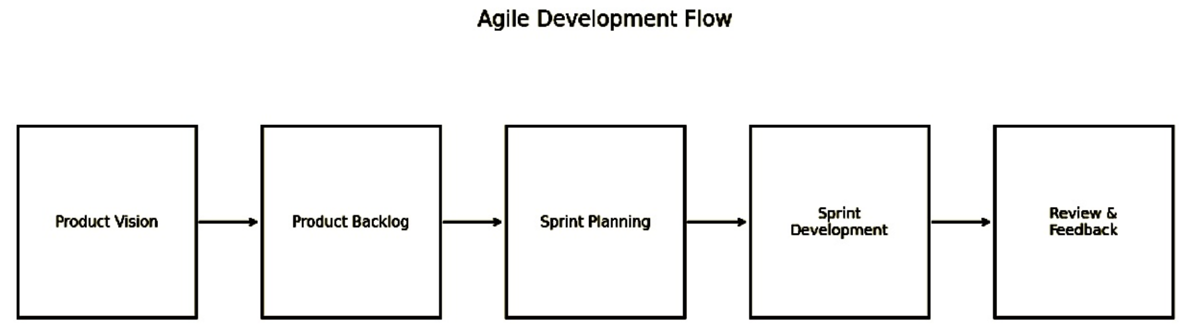

Figure 1. Agile flow for the mini-project.

4. Stakeholders and Users

The source requires the team to study stakeholders and user personas. Since the supplied Unit–2 document does not name specific personas, the following mapping is a proposed project artifact derived from the listed capabilities.

Stakeholder/User	Interest	Typical Needs
Student	Academic information	View profile, courses, attendance, results and notifications.
Faculty Member	Academic administration	Record attendance, enter marks, manage courses and generate reports.
Institution/Administration	Operational oversight	Reliable records, reporting and access control.
Product Owner	Product value	Prioritize requirements and accept completed work.
Scrum Master	Process effectiveness	Facilitate Scrum, remove impediments and support collaboration.

Development Team	Product delivery	Design, develop, test and document the product.

5. Scrum Team Structure and Roles

The team organizes itself as a Scrum Team and documents responsibilities for each role.

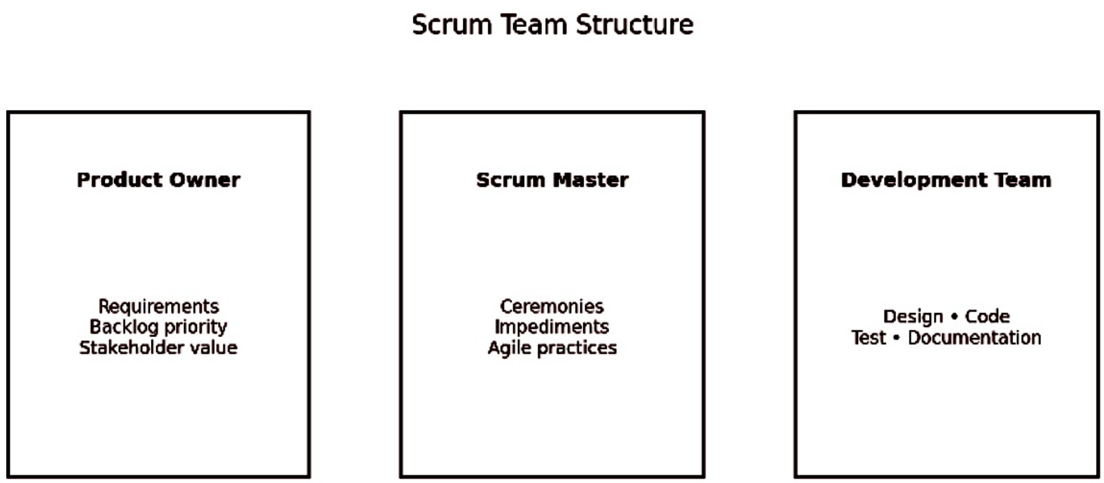

Figure 2. Scrum team structure.

Role	Responsibilities
Product Owner	Define requirements; prioritize Product Backlog; represent stakeholders; accept completed work.
Scrum Master	Facilitate ceremonies; remove impediments; ensure Agile practices; support collaboration.
Development Team	Design application; write code; test software; prepare documentation; demonstrate features.

6. Project Epics and Feature Scope

The application is decomposed into seven high-level Epics. The feature lists below follow the supplied Unit–2 brief.

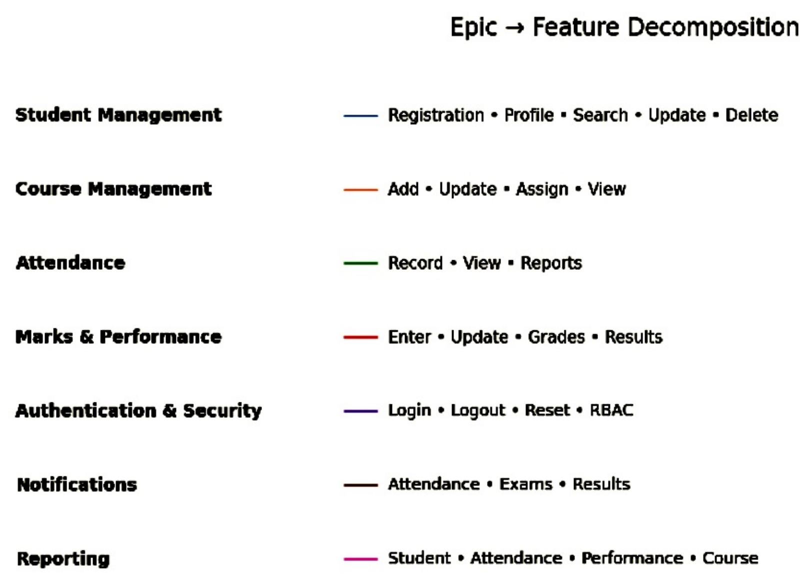

Figure 3. Epic-to-feature decomposition.

Epic	Features
Epic 1 – Student Management	Student Registration; Profile Management; Search; Update; Deletion
Epic 2 – Course Management	Add Course; Update Course; Assign Students; View Course Details
Epic 3 – Attendance Management	Record Attendance; View Attendance; Attendance Reports
Epic 4 – Marks and Academic Performance	Enter Marks; Update Marks; Calculate Grades; Generate Result Sheets
Epic 5 – Authentication and Security	Login; Logout; Password Reset; Role-Based Access Control
Epic 6 – Notifications	Attendance Alerts; Examination Notifications; Result Notifications
Epic 7 – Reporting	Student Reports; Attendance Reports; Performance Reports; Course Reports

7. User Stories

User Stories break Epics into user-focused requirements using the format: As a [user], I want to [action] so that [benefit].

Epic	User Story	Priority
Authentication	As an authorized user, I want to log in securely so that I can access permitted functions.	High
Student Management	As an authorized user, I want to add a student so that a current	High

	student record is maintained.	
Student Management	As an authorized user, I want to search for a student so that academic information can be found quickly.	High
Student Management	As an authorized user, I want to update a student profile so that records remain accurate.	High
Student Management	As an authorized user, I want to delete a permitted student record so that obsolete data can be removed.	Medium
Course Management	As a faculty member, I want to add and update courses so that course information remains current.	Medium
Course Management	As an authorized user, I want to assign students to courses so that enrollment is maintained.	Medium
Attendance	As a Faculty Member, I want to record student attendance so that attendance records remain accurate and updated.	High
Attendance	As a user, I want to view attendance so that attendance status can be monitored.	Medium
Marks	As a Faculty Member, I want to enter and update marks so that academic performance is maintained.	Medium
Marks	As an authorized user, I want grades and result sheets generated so that academic outcomes can be communicated.	Medium
Notifications	As a student, I want attendance, examination and result notifications so that I do not miss important updates.	Medium
Reporting	As an administrator or faculty member, I want reports so that academic information can be reviewed and communicated.	Medium

8. Acceptance Criteria and Definition of Done

Acceptance Criteria are measurable conditions that determine whether a User Story is complete. The source specifically states that they provide a clear definition of Done.

8.1 Student Login – Acceptance Criteria

Login page loads successfully.

Registered users can log in.

Invalid credentials show an appropriate error message.

Successful login redirects to the dashboard.

Session management functions correctly.

8.2 Record Attendance – Acceptance Criteria

Faculty can select a course.

Faculty can select a lecture date.

Faculty can mark attendance.

Attendance records are saved successfully.

Attendance reports can be generated.

8.3 Proposed Definition of Done

Implementation is complete.

All Acceptance Criteria are satisfied.

Relevant testing is completed.

No known blocking defect remains.

Documentation is updated where required.

Completed work is demonstrated and accepted in Sprint Review.

9. Product Backlog and Prioritization

Backlog prioritization considers business value, user importance, technical dependencies and implementation complexity.

Order	Epic	User Story	Priority
1	Authentication	User Login	High
2	Student Management	Add Student	High
3	Student Management	Search Student	High
4	Attendance	Record Attendance	High
5	Course Management	Add Course	Medium
6	Reporting	Attendance Report	Medium

7	Notifications	Attendance Alert	Medium

The first four priorities reproduce the examples in the supplied brief. Remaining entries are organized using the same prioritization logic and should be refined by the Product Owner.

10. Story Point Estimation

Story Points represent relative effort and complexity. The supplied estimation scale is:

Points	Complexity
1	Very Easy
2	Easy
3	Moderate
5	Difficult
8	Very Difficult
13	Highly Complex

Planning Poker can be used for collaborative estimation. The following values are illustrative planning estimates, not actual completed measurements.

Story	Epic	Illustrative Points
User Login	Authentication	3
Add Student	Student Management	3
Search Student	Student Management	2
Record Attendance	Attendance	5
Add Course	Course Management	3
Attendance Report	Reporting	5
Attendance Alert	Notifications	5
Academic Analytics Dashboard	Dashboard	8

11. Sprint Planning and Sprint Backlogs

The supplied brief defines three development Sprints.

Sprint	Goal	Activities
Sprint 1	User authentication and project setup	Repository initialization; user registration; login; basic UI; testing.
Sprint 2	Student Management Module	Add; Search; Update; Delete Student.
Sprint 3	Course and Attendance Management	Add Course; Assign Courses; Record Attendance; Attendance Reports.

11.1 Expected Sprint Deliverables

Sprint 1: working foundation, registration/login flow, basic interface and tests.

Sprint 2: core student-management operations.

Sprint 3: course assignment, attendance recording and attendance reporting.

12. Scrum Board and Workflow

The Scrum Board tracks work through Backlog → To Do → In Progress → Testing → Completed.

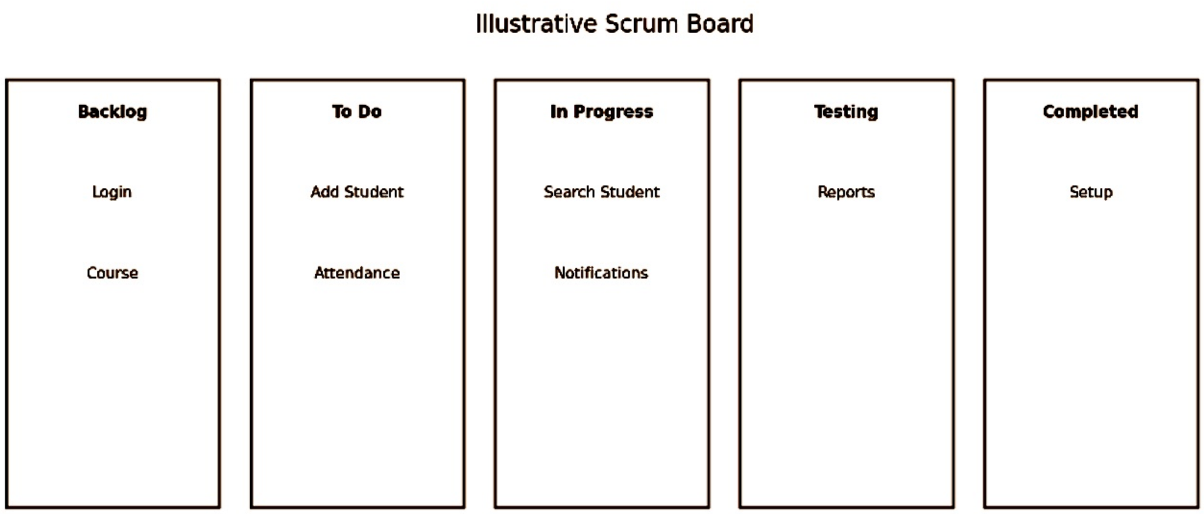

Figure 4. Illustrative Scrum Board based on the source workflow.

13. Scrum Ceremonies

13.1 Sprint Planning

Select User Stories, estimate effort and define the Sprint Goal.

13.2 Daily Scrum

What did I complete yesterday?

What will I do today?

What challenges am I facing?

13.3 Sprint Review

Demonstrate completed features and collect stakeholder feedback.

13.4 Sprint Retrospective

What went well?

What challenges occurred?

What improvements should be implemented?

14. Conceptual Product Workflow

This conceptual view connects the major functional areas from the seven Epics. It is a functional relationship diagram, not a technical architecture.

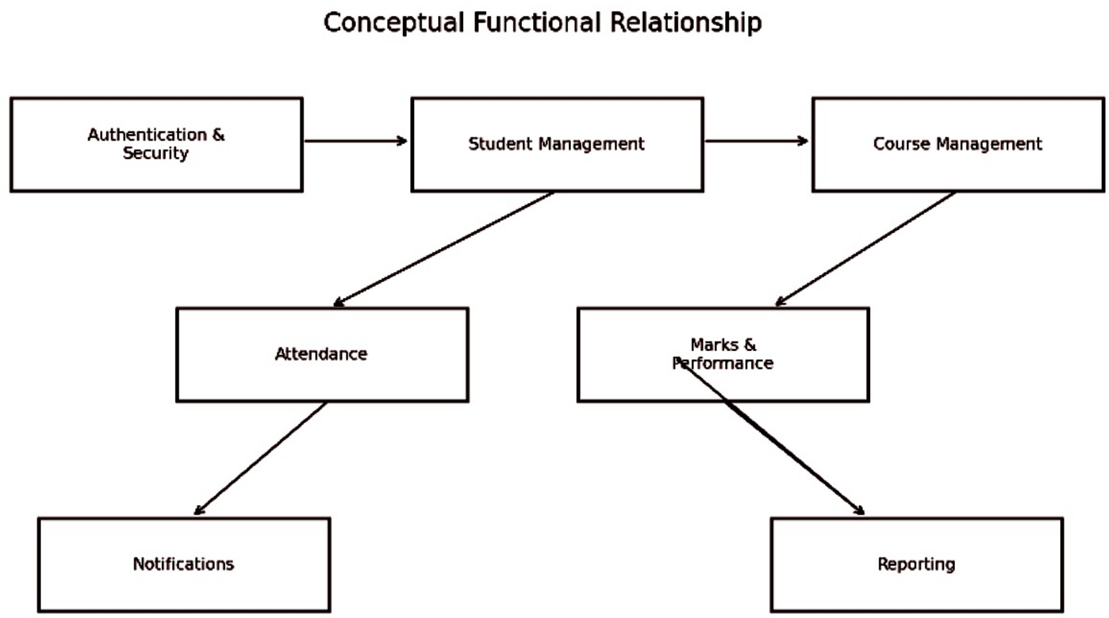

Figure 5. Conceptual relationship among the major platform modules.

15. Testing and Quality Considerations

Testing is included in the Sprint plan and in Development Team responsibilities. The following checklist converts the Acceptance Criteria approach into practical verification areas.

Area	Verification
Authentication	Valid login succeeds; invalid credentials are rejected; session behavior works.
Student Management	Add, search, update and delete operations satisfy their criteria and permissions.
Course Management	Courses can be added/updated and students assigned.
Attendance	Course/date selection, marking and saving are verified.
Marks	Marks are entered/updated and grades/results are generated.
Notifications	Relevant academic events trigger intended notifications.
Reporting	Required reports can be generated and contain expected information.

Security	Role-based access prevents unauthorized actions.

Actual test results, defects and pass/fail evidence should be added during implementation; the supplied brief does not provide completed test data.

16. Agile Framework Comparison

The brief requires comparison of Scrum, Kanban, XP, Lean Development, LeSS, SAFe and Scrum@Scale.

Framework	Core Idea	Suitability for This Mini-Project
Scrum	Time-boxed Sprints, defined roles and ceremonies	Strong fit; aligns directly with the supplied plan.
Kanban	Continuous flow and visual work management	Possible alternative where continuous flow is preferred.
Extreme Programming (XP)	Engineering practices and frequent feedback	Useful complementary practices for software quality.
Lean Development	Reduce waste and maximize value	Useful principles for efficient delivery.
LeSS	Scale Scrum across multiple teams	Likely unnecessary for a small academic team.
SAFe	Enterprise-scale Agile	Too heavy for this mini-project.
Scrum@Scale	Coordinate multiple Scrum teams	Likely unnecessary unless the project expands significantly.

16.1 Recommendation

Scrum is the most suitable primary framework because the project already defines a Product Owner, Scrum Master, Development Team, Product Backlog, Sprints, Sprint Planning, Daily Scrum, Sprint Review and Sprint Retrospective. Enterprise scaling frameworks would add unnecessary overhead for the stated project size.

17. Release Roadmap and Timeline

Phase	Outcome	Key Deliverables
Preparation	Agile setup	Product Vision; Scrum roles; Epics; initial Product Backlog
Sprint 1	Foundation	Repository; registration; login; basic UI; tests
Sprint 2	Student management	Add/search/update/delete student
Sprint 3	Course & attendance	Course management; assignment; attendance; reports
Review/Release	Integrated increment	Demo; feedback; retrospective; roadmap refinement

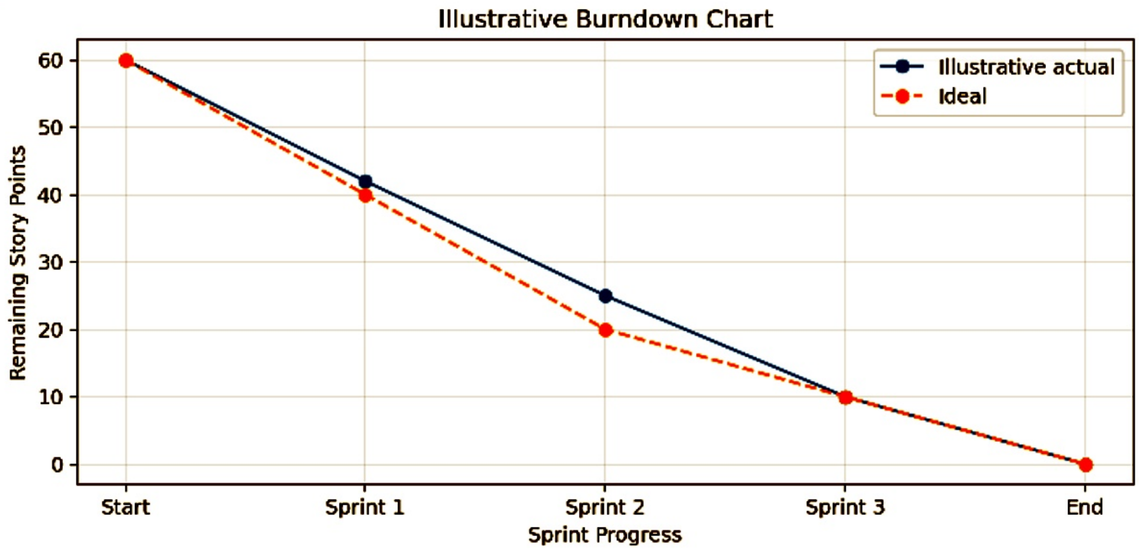

Figure 6. Illustrative burndown chart; values are examples, not actual project measurements.

18. Risks, Constraints and Mitigation

Risk/Constraint	Impact	Mitigation
Unclear requirements	Scope changes	Use Acceptance Criteria, backlog refinement and Product Owner prioritization.
Technical dependencies	Stories may be blocked	Prioritize foundational stories and identify dependencies in Sprint Planning.
Limited team capacity	Sprint overload	Use Story Points and refine capacity after initial Sprints.
Integration defects	Modules may fail together	Integrate incrementally and test each Sprint.
Security issues	Unauthorized access	Apply authentication and role-based access control; test authorization.
Incomplete documentation	Required submissions may be missed	Maintain a deliverables checklist throughout the project.

19. Expected Outcomes and Success Criteria

High-priority authentication, student and attendance capabilities are implemented and accepted.

User Stories satisfy their Acceptance Criteria.

Sprint Goals are defined and reviewed.

Scrum Board reflects actual work status.

Sprint Reviews capture stakeholder feedback.

Retrospective findings lead to concrete improvements.

Agile documentation remains complete and consistent.

The project demonstrates incremental delivery across planned Sprints.

20. Conclusion

The Agile mini-project transforms the Student Record and Academic Management Platform into an actionable development plan. Seven Epics are translated into User Stories, prioritized in a Product Backlog, estimated with Story Points and organized across three Sprints. Scrum roles and ceremonies support collaboration, while Acceptance Criteria establish measurable completion conditions.

The report can serve as the planning foundation for implementation and academic evaluation. During development, illustrative estimates, board entries, burndown values and proposed stakeholder details should be replaced with actual project evidence.

Appendix A – Complete Agile Documentation Checklist

1. Agile Project Planning Report
2. Product Vision Document
3. Scrum Team Structure
4. Scrum Roles and Responsibilities Document
5. Epic Documentation
6. User Story Repository
7. Acceptance Criteria Document
8. Prioritized Product Backlog
9. Story Point Estimation Sheet
10. Sprint Planning Document
11. Sprint Backlog
12. Scrum Board
13. Burndown Chart
14. Sprint Review Report
15. Sprint Retrospective Report
16. Agile Framework Comparison Report
17. Project Timeline and Release Roadmap
18. Complete Agile Documentation

Source basis: supplied “UNIT-2 AGILE MINI-PROJECT” document. Illustrative artifacts are explicitly labeled where the source did not provide actual project data.